

SaaS Subscription Churn Prediction & Customer Retention Analysis

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Tools Used: Power BI, Power Query, DAX, Microsoft Excel

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1. Executive Summary

Customer retention is one of the most important factors influencing the long-term profitability of subscription-based businesses. High customer churn leads to reduced recurring revenue, increased customer acquisition costs, and slower business growth.

This case study analyzes customer subscription data to identify the primary drivers of churn and recommend strategies to improve customer retention. Using Power BI, the data was cleaned, transformed, and visualized through interactive dashboards to evaluate customer behavior across contract types, payment methods, service subscriptions, tenure, and pricing.

The analysis identified several high-risk customer segments, including customers on month-to-month contracts, Electronic Check users, Fiber Optic subscribers, customers without Tech Support, and newly acquired customers. Based on these findings, targeted business recommendations have been proposed to reduce churn and improve customer lifetime value.

2. Business Problem

A subscription-based SaaS company has observed an increase in customer cancellations, resulting in reduced recurring revenue and lower customer lifetime value. The management team seeks to understand the underlying causes of customer churn and identify actionable strategies to improve customer retention.

The objective of this analysis is to examine customer demographics, subscription details, service usage, billing preferences, and tenure to identify the factors that contribute most significantly to customer attrition.

3. Business Objectives

- The primary objectives of this analysis were:
- Measure the overall customer churn rate.
 - Identify customer segments with the highest churn.
 - Analyze the impact of contract type, payment method, pricing, and service adoption on churn.
 - Evaluate customer behavior across different subscription plans.
 - Develop actionable business recommendations to improve customer retention and reduce revenue loss.
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4. Dataset Overview

The analysis was performed using a publicly available customer churn dataset containing information on **7,000+ subscription customers**.

Dataset Characteristics

Attribute	Description
Total Customers	~7,000
Customer Demographics	Gender, Partner, Dependents, Senior Citizen
Subscription Details	Contract Type, Internet Service, Monthly Charges

Attribute	Description
Service Features	Tech Support, Online Security, Device Protection
Billing Information	Payment Method, Paperless Billing
Target Variable	Customer Churn (Yes/No)

The dataset was used as a simulated SaaS subscription business to demonstrate business analysis techniques.

5. Methodology

The analysis followed a structured business analytics process.

Data Preparation

- Imported customer data into Power BI.
- Verified data types.
- Removed incomplete records.
- Created business-friendly calculated columns using DAX.
- Built measures for customer KPIs including:
 - Total Customers
 - Churned Customers
 - Churn Rate
 - Total Revenue
 - Average Monthly Charges
 - Average Tenure

Dashboard Development

Two interactive dashboards were created:

- Executive Overview
- Customer Behavior Analysis

The dashboards allow business users to interactively analyze churn across multiple customer segments using slicers and filters.

6. Dashboard Overview

Dashboard 1 – Executive Overview

The Executive Dashboard provides a high-level summary of customer churn through key performance indicators and interactive visualizations.

The dashboard enables decision-makers to quickly evaluate customer retention performance, contract distribution, customer segmentation, and service usage patterns.

Dashboard 2 – Customer Behavior Analysis

The Customer Analysis Dashboard focuses on identifying the underlying drivers of customer churn.

It evaluates churn across contract types, payment methods, internet services, technical support availability, customer tenure, and monthly pricing bands, enabling management to identify high-risk customer segments.

7. Key Findings

Finding 1 – Contract Type

Customers subscribed to **month-to-month contracts** exhibit the highest churn rate, indicating that short-term subscription plans are associated with increased customer attrition.

Finding 2 – Payment Method

Customers using **Electronic Check** as their preferred payment method demonstrate the highest churn rate, suggesting that payment preferences may influence customer retention.

Finding 3 – Technical Support

Customers without **Tech Support** experience significantly higher churn than customers who subscribe to technical support services, highlighting the importance of customer assistance in improving retention.

Finding 4 – Internet Service

Customers using **Fiber Optic Internet Service** exhibit the highest churn rate among all service categories, indicating a need for further investigation into pricing, service quality, or customer expectations.

Finding 5 – Customer Lifecycle

Customers classified as **New Customers** and those within the **High Monthly Charge Band** experience the highest churn rates, suggesting that both onboarding experience and pricing strategies influence customer retention.

8. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Encourage month-to-month customers to migrate to annual subscription plans through loyalty discounts and promotional offers.
 - Investigate the payment experience for Electronic Check users and encourage alternative payment methods where appropriate.
 - Increase adoption of Tech Support by bundling it with subscription plans or offering introductory support packages.
 - Conduct customer satisfaction surveys for Fiber Optic subscribers to identify service quality concerns.
 - Develop targeted onboarding programs and personalized engagement campaigns for new customers.
 - Introducing retention incentives for customers on higher-priced subscription plans to reduce early churn.
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9. Conclusion

This analysis demonstrates how business intelligence tools can be used to identify customer churn patterns and generate actionable business insights. Using Power BI dashboards, DAX calculations, and interactive visualizations, the primary drivers of customer attrition were identified.

The findings suggest that customer retention can be improved through targeted interventions focusing on contract optimization, customer support services, pricing strategies, and

onboarding initiatives. Implementing these recommendations has the potential to reduce churn, improve customer satisfaction, and increase long-term customer lifetime value.